4.2.8 PostScript

Mathematica generates all its graphics output in PostScript. The two-dimensional graphics primitives that *Mathematica* uses are quite similar to those in PostScript.

PostScript	*Mathematica*
`gsave` *primitives* `grestore`	`{` *primitives* `}`
x y `moveto 0 0 rlineto stroke`	`Point[{`x`, `y`}]`
x_1 y_1 `moveto` x_2 y_2 `lineto ... stroke`	`Line[{{`x_1`, `y_1`}, {`x_2`, `y_2`}, ...}]`
`newpath` x_1 y_1 `moveto` x_2 y_2 `moveto ... fill`	`Polygon[{{`x_1`, `y_1`}, {`x_2`, `y_2`}, ...}]`
`... image`	`CellArray[{{...}, ...}]`
dx `setlinewidth`	`Thickness[`*dx*`]`
level `setgray`	`GrayLevel[`*level*`]`
r g b `setrgbcolor`	`RGBColor[`r`, `g`, `b`]`
`... setdash`	`Dashing[{...}]`

Some correspondences between PostScript and *Mathematica* `Graphics` primitives.

The choice of coordinate system in *Mathematica* is specified by the option `PlotRange`. It is often based on the actual objects you are plotting.

Most *Mathematica* graphics is specified at a much higher level than PostScript. So, for example, the option `Axes` determines whether a complete set of axes should be drawn. In addition, when you plot a function using `Plot`, *Mathematica* chooses sample points using an adaptive algorithm, so it does not have to resort to splines to get a smooth curve. (`ContourPlot` nevertheless does use Bézier curves for contour lines.)

Mathematica primitives such as `Thickness` and `Dashing` specify distances as fractions of the width of the whole plot. The *Mathematica* primitive `PointSize[`*size*`]` specifies the diameter of a single dot or point in these units.

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; http://www.wolfram.com/; 1-800-441-6284.